

Hexa HX-400 Heavy Haul Tractor

Operator & Fleet Maintenance Manual

Sample document: this is a fictional vehicle created to demonstrate multi-model retrieval in the Fleet Assistance Copilot. Values are realistic but not taken from any real OEM.

Document Control

Field	Value
Document ID	VM-HX400-OM
Vehicle model	HX-400
Variants covered	HX-400 4x2, HX-400 6x4
Vehicle category	N3 heavy goods tractor (diesel, BS-VI)
Region	India
Version	1.2
Effective date	2026-07-01

1. About This Manual

This manual covers operation, preventive maintenance and telematics thresholds specific to the Hexa HX-400 heavy haul tractor. Where this manual and the Fleet Operations & Driver Safety Handbook differ, the values in this manual take priority for HX-400 vehicles because they are calibrated to this model.

The HX-400 is used for long-haul full-truck-load and container movements. Both variants share the same engine and cab; the 6x4 variant adds a tandem drive axle, a hydraulic retarder and a higher gross combination weight.

2. Vehicle Specifications

Key technical specifications for both HX-400 variants are listed below.

Table 2.1 – HX-400 technical specifications

Specification	HX-400 4x2	HX-400 6x4
Engine	7.2 L inline-6 diesel, BS-VI	7.2 L inline-6 diesel, BS-VI
Maximum power	400 hp @ 1,900 rpm	400 hp @ 1,900 rpm
Maximum torque	1,850 Nm @ 1,000–1,400 rpm	1,850 Nm @ 1,000–1,400 rpm
Transmission	12-speed AMT	12-speed AMT
Gross combination weight (GCW)	40,000 kg	49,000 kg
Fuel tank capacity	415 L	415 L
AdBlue (DEF) tank capacity	60 L	60 L
Speed governor setting	80 km/h	80 km/h
Service brakes	Dual-circuit full air, S-cam, ABS	Dual-circuit full air, S-cam, ABS
Auxiliary braking	Engine exhaust brake	Engine exhaust brake + hydraulic retarder
Tyre size	295/80 R22.5	295/80 R22.5

3. Fluid and Consumable Capacities

Table 3.1 – HX-400 fluid capacities

System	Capacity	Specification
Engine oil (with filter)	28 L	SAE 15W-40, API CK-4
Engine coolant	45 L	OAT coolant, 50:50 premix
AMT transmission oil	12 L	SAE 75W-80 synthetic gear oil
Rear axle oil (4x2)	16 L	SAE 85W-140, API GL-5
Rear axle oil (6x4, per axle)	14 L	SAE 85W-140, API GL-5
Power steering fluid	5 L	ATF Dexron III
Diesel exhaust fluid (AdBlue)	60 L	ISO 22241 compliant AdBlue

4. Tyre Specifications and Pressures

Check cold tyre pressures before every trip. Hot pressures after a long run are normally 0.5–0.8 bar higher; never bleed air from hot tyres.

Table 4.1 – HX-400 tyre pressures

Position	Cold pressure	Minimum tread depth (company policy)
Steer axle	8.3 bar (120 psi)	3.0 mm
Drive axle(s)	7.6 bar (110 psi)	2.0 mm
Trailer axles	8.3 bar (120 psi)	2.0 mm

Retreaded tyres must not be fitted on the steer axle of an HX-400. The legal minimum tread depth is 1.6 mm, but company policy requires replacement at the depths above.

5. Instrument Cluster Warning Lights

Red lamps mean stop safely as soon as possible. Amber lamps mean the vehicle can continue but must be inspected soon. Green lamps are informational.

Table 5.1 – HX-400 warning lamps

Warning lamp	Colour	Meaning	Driver action
Low air pressure	Red	System air pressure below 5.5 bar; buzzer sounds	Stop safely immediately. Do not drive until pressure is above 6.5 bar.
Engine coolant temperature	Red	Coolant above 108 °C	Stop, idle 2–3 minutes, then switch off. Do not open the radiator cap when hot.
Engine oil pressure	Red	Oil pressure below safe limit	Stop and switch off the engine immediately.
Check engine (MIL)	Amber	Engine or emissions fault stored	Continue with care; book inspection within 24 hours.
DPF regeneration required	Amber	Soot load high in diesel particulate filter	Perform parked regeneration (30–45 minutes) at a safe location.
AdBlue (DEF) low	Amber	DEF level below 10%	Refill AdBlue. Engine torque is derated when the tank is empty.
ABS fault	Amber	ABS disabled; base brakes still work	Drive gently, avoid hard braking, book inspection.
Water in fuel	Amber	Water detected in fuel pre-filter	Drain the water separator at the next stop.

Warning lamp	Colour	Meaning	Driver action
Exhaust brake active	Green	Exhaust brake engaged	Information only.

6. Common Diagnostic Trouble Codes (DTC)

Table 6.1 – Frequent HX-400 DTCs

DTC	Description	Severity	Recommended action
P0217	Engine over-temperature condition	Critical	Stop safely, check coolant level and fan belt; tow to workshop if it recurs.
P0524	Engine oil pressure too low	Critical	Switch off immediately; do not restart until oil level and pressure are checked.
P2463	DPF soot accumulation excessive	High	Run parked regeneration; if it fails, visit workshop within 1 day.
P20EE	SCR NOx catalyst efficiency below threshold	Medium	Check AdBlue quality and level; book emissions inspection.
P0087	Fuel rail pressure too low	High	Check fuel level and filters; replace fuel filters if clogged.
C0035	ABS wheel speed sensor circuit fault	Medium	Inspect sensor wiring; ABS may be disabled.
U0100	Lost communication with engine control module	High	Check CAN wiring and telematics unit; inspect at workshop.

7. Preventive Maintenance Schedule

Service at the kilometre or time interval, whichever comes first. For severe duty (mining, ghat sections, more than 20% idling) halve the engine oil interval.

Table 7.1 – HX-400 preventive maintenance schedule

Interval	Service items
Daily (pre-trip)	Walk-around, tyre pressure, air tank drain, fluid leaks, lights, fifth-wheel lock, AdBlue level
Every 10,000 km	Chassis greasing, fifth-wheel greasing, air dryer check, brake stroke check
40,000 km or 6 months	Engine oil and oil filter change, fuel filters, air filter inspection, DPF differential-pressure check
80,000 km or 12 months	Coolant concentration test, transmission oil level check, brake lining measurement, AdBlue filter replacement
160,000 km or 24 months	Coolant replacement, transmission oil replacement, axle oil replacement, valve clearance adjustment

8. Telematics Thresholds for HX-400

These thresholds are configured in the telematics platform for all HX-400 vehicles. They are tighter than the generic handbook ranges because a laden 40-tonne combination needs much longer stopping distances.

Table 8.1 – HX-400 telematics thresholds

Signal / event	Warning threshold	Critical threshold
Harsh braking	Deceleration ≥ 0.30 g	Deceleration ≥ 0.45 g
Harsh acceleration	Acceleration ≥ 0.25 g	—
Harsh cornering	Lateral ≥ 0.28 g	Lateral ≥ 0.35 g (rollover risk)

Signal / event	Warning threshold	Critical threshold
Overspeed	> 80 km/h for 10 s	> 85 km/h (governor tamper suspected)
Excessive idling	> 5 min continuous	> 20% of engine-on time per day
Over-revving	> 2,100 rpm	> 2,300 rpm
Coolant temperature	≥ 105 °C	≥ 108 °C
System air pressure	< 6.5 bar	< 5.5 bar
AdBlue level	< 10%	0% (torque derate)

9. Fuel Efficiency Guidance

Keep engine speed in the green band of 1,000–1,500 rpm on the tachometer. The AMT selects gears automatically, but use Eco mode on highways and avoid manual downshifts except on descents.

Table 9.1 – HX-400 expected fuel economy

Load condition	Expected fuel economy	Notes
Empty tractor-trailer	4.2–4.8 km/L	Highway, 60–70 km/h cruise
Half load (~20 t GCW)	3.4–3.8 km/L	Mixed highway
Full load (40 t GCW)	2.6–3.0 km/L	Flat highway; ghat sections lower

- Use cruise control on flat highways at 60–70 km/h.
- Anticipate traffic and coast in gear instead of braking late.
- Keep tyres at the specified cold pressure; 1 bar under-inflation adds about 2% fuel use.
- Switch off the engine when idling for more than 5 minutes.

10. Trailer Coupling and Load Limits

1. Apply the trailer parking brake and chock the trailer wheels.
2. Check that the fifth-wheel jaws are open and the plate is greased.
3. Reverse slowly until the kingpin locks; you will hear a distinct click.
4. Perform a tug test in low gear with the trailer brakes applied.
5. Visually confirm the lock indicator and safety latch are engaged.
6. Connect air lines (red = emergency, yellow = service) and the electrical connector.
7. Raise the landing gear fully and remove chocks before moving.

Table 10.1 – HX-400 load limits

Axle / limit	HX-400 4x2	HX-400 6x4
Front axle load limit	7,000 kg	7,500 kg
Rear axle load limit	11,500 kg	2 × 10,200 kg (tandem)
Fifth-wheel load limit	16,000 kg	20,000 kg
Maximum GCW	40,000 kg	49,000 kg

11. Emergency Procedures

11.1 Low Air Pressure

If the red low-air lamp and buzzer come on, stop safely at once. Below about 4.5 bar the spring brakes apply automatically and the vehicle will stop on its own. Never try to drive with the spring brakes caged.

11.2 Brake Fade on Long Descents

Select a low gear before the descent, keep the exhaust brake (and retarder on 6x4) engaged, and use the service brakes in firm, short applications. If braking feels weak, reduce speed early and use an escape ramp if available.

11.3 DPF Regeneration

A parked regeneration takes 30–45 minutes. Park away from dry grass and fuel, keep the engine running and do not switch off during regeneration. Exhaust temperatures exceed 600 °C.

12. Frequently Asked Questions

Q: How often should I change the engine oil on the HX-400? A: Every 40,000 km or 6 months, whichever comes first; every 20,000 km on severe duty.

Q: What happens if AdBlue runs out? A: The engine control unit derates torque until AdBlue is refilled. Keep the level above 10%.

Q: At what coolant temperature should I stop? A: Stop when the red coolant lamp comes on at 108 °C; a telematics warning is raised at 105 °C.